Variational Inference based on Robust Divergences

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Task/Model Motivation

- Real-world data contains outliers, but standard methods fail
  Ordinary variational inference treats all data points equally, making it highly sensitive to outliers—even a single contaminated point can arbitrarily bias the model
- Existing robust approaches don't scale to complex models
  Traditional model-based robustness (e.g., replacing Gaussian with Student-t likelihood) only works for simple statistical models
- Variational inference needs robustness without sacrificing flexibility
  There is a critical need for a principled framework that maintains the advantages of VI (speed, scalability) while providing mathematically proven protection against outliers

Formal Problem Statement

Given:

- Dataset $\{x_i\}_{i=1}^N$ with potential outliers (contamination rate ϵ)
- Parametric model $p(x|\theta)$ with prior $p(\theta)$
- Empirical distribution $\hat{p}(x) = \frac{1}{N}\sum_{i=1}^N \delta(x, x_i)$

Classical Variational Inference:

$$q^*(\theta) = \arg\min_{q(\theta)\in\mathcal{Q}} \underbrace{D_{\mathrm{KL}}(q(\theta)\|p(\theta))}_{\text{regularization}} - \int q(\theta) \underbrace{(-N d_{\mathrm{KL}}(\hat{p}(x)\|p(x|\theta)))}_{\text{data fitting}} d\theta$$

Goal: Design robust VI with bounded influence while maintaining:

- Scalability to complex models (deep networks)
- Computational efficiency of variational methods

Variational Inference: Core Idea

Bayesian Inference Goal: Compute posterior $p(\theta|x_{1:N})$

$$p(\theta|x_{1:N}) = \frac{p(x_{1:N}|\theta)p(\theta)}{p(x_{1:N})} = \frac{p(x_{1:N}|\theta)p(\theta)}{\int p(x_{1:N}|\theta)p(\theta)d\theta}$$

Problem: Integral intractable for complex models!

VI Solution: Approximate $p(\theta|x_{1:N})$ with $q(\theta;\lambda) \in \mathcal{Q}$

Optimization Formulation (Zellner, 1988)

$$q^*(\theta) = \arg\min_{q\in\mathcal{Q}} \underbrace{D_{\mathrm{KL}}(q(\theta)\|p(\theta))}_{\text{stay close to prior}} + \underbrace{\mathbb{E}_q[D_{\mathrm{KL}}(\hat{p}(x)\|p(x|\theta))]}_{\text{fit data}}$$

Key Properties:

- Converts inference $\rightarrow$ optimization problem (faster than MCMC)
- Maximizes Evidence Lower BOund (ELBO):

Standard Variational Inference: Two Main Approaches

1. Mean-Field Variational Bayes (MFVB)

Assume factorized posterior: $q(\theta) = \prod_{j=1}^{k} q_j(\theta_j)$

Algorithm: Coordinate ascent, update each factor iteratively

$$q_j(\theta_j) \propto \exp\left(\mathbb{E}_{q_{-j}}[\log p(x, \theta)]\right)$$

- ✓ Analytical solution for conjugate priors
- × Ignores posterior dependencies

2. Fixed-Form Variational Bayes (FFVB / Black-Box VI)

Fix parametric family: $q(\theta; \lambda)$ (e.g., Gaussian $\mathcal{N}(\mu, \Sigma)$)

Algorithm: Gradient descent on λ

$$\nabla_\lambda \mathcal{L} = \mathbb{E}_{q_\lambda}[\nabla_\lambda \log q_\lambda(\theta) \cdot (\log p(x, \theta) - \log q_\lambda(\theta))]$$

Techniques: Reparameterization trick, control variates, natural gradients

- ✓ Flexible, applies to deep networks
- × Requires careful variance reduction

Limitation of Standard VI → Robust Divergences

Problem with KL Divergence:

Standard VI uses cross-entropy for data fitting:

$$d_{\mathrm{KL}}(\hat{p}(x)\|p(x|\theta)) = -\frac{1}{N}\sum_{i=1}^{N}\ln p(x_i|\theta)$$

All data points weighted equally → outliers dominate!

β-**Divergence** (Basu et al., 1998)

$$D_\beta(g\|f) = \frac{1}{\beta}\int g^{1+\beta}(x)dx - \frac{\beta+1}{\beta}\int g(x)f^{\beta}(x)dx + \int f^{1+\beta}(x)dx$$

Key Property:

$$\lim_{\beta\to 0} D_\beta(g\|f) = \lim_{\gamma\to 0} D_\gamma(g\|f) = D_{\mathrm{KL}}(g\|f)$$

Idea: Replace KL with robust divergence in **data-fitting term only!**

β-Variational Inference: Mathematical Framework

Modified Objective (β-ELBO):

$$\mathcal{L}_\beta(q(\theta)) = \underbrace{D_{\mathrm{KL}}(q(\theta)\|p(\theta))}_{\text{keep KL for prior!}} - \int q(\theta)\underbrace{\left(-Nd_\beta(\hat{p}(x)\|p(x|\theta))\right)}_{\text{robust data fitting}} d\theta$$

β-Cross-Entropy:

$$d_\beta(\hat{p}(x)\|p(x|\theta)) = -\frac{\beta+1}{\beta}\frac{1}{N}\sum_{i=1}^{N}\underbrace{p(x_i|\theta)^\beta}_{\text{down-weight outliers}} + \int p(x|\theta)^{1+\beta}dx$$

Theorem 1: Pseudo-Posterior

Solution of $\arg\min_q \mathcal{L}_\beta(q)$ is:

$$q^*(\theta) = \frac{e^{-Nd_\beta(\hat{p}(x)\|p(x|\theta))}p(\theta)}{\int e^{-Nd_\beta(\hat{p}(x)\|p(x|\theta))}p(\theta)d\theta}$$

Theoretical Guarantees: Influence Function Analysis

Influence Function (IF): Measures sensitivity to contamination

$$\mathrm{IF}(z, T, G) = \lim_{\epsilon \to 0} \frac{T((1-\epsilon)G + \epsilon\delta_z) - T(G)}{\epsilon}$$

Robustness criterion: $\sup_z |\mathrm{IF}(z, \theta, G)| < \infty$ (bounded)

Theorem 2: IF for VI

Ordinary VI:

$$\mathrm{IF}(z, m^*, G) = \left(\frac{\partial^2 \mathcal{L}}{\partial m^2}\right)^{-1} \frac{\partial}{\partial m} \mathbb{E}_{q^*}\left[D_{\mathrm{KL}}(q^* \| p) + N \ln p(z|\theta)\right]$$

β-**VI:**

$$\mathrm{IF}(z, m^*, G) = \left(\frac{\partial^2 \mathcal{L}_\beta}{\partial m^2}\right)^{-1} \frac{\partial}{\partial m} \mathbb{E}_{q^*}\left[D_{\mathrm{KL}}(q^* \| p) + N I_\beta(z)\right]$$

where $I_\beta(z) = \frac{\beta+1}{\beta} p(z|\theta)^\beta - \int p(x|\theta)^{1+\beta} dx$ is bounded!

Why Does β-VI Have Bounded Influence?

Ordinary VI:

$$\text{weight}(x_i) = 1$$

- All points equally weighted
- For ReLU: as $x_o \to \infty$, activation grows unbounded
- IF diverges $\to$ no robustness

β-VI:

$$\text{weight}(x_i) = p(x_i|\theta)^\beta$$

- Outliers downweighted
- For ReLU: $p(x_o|\theta) \to 0$ $\Rightarrow$ weight $\to 0$ exponentially
- IF bounded $\to$ **robust**

Activation function	Regression	β- and γ-Regression	Classification	β- and γ-Classification
Linear	$(x_o : U, y_o : U)$	$(x_o : B, y_o : B)$	$(x_o : U)$	$(x_o : B)$
ReLU	$(x_o : U, y_o : U)$	$(x_o : B, y_o : B)$	$(x_o : U)$	$(x_o : B)$
tanh	$(x_o : B, y_o : U)$	$(x_o : B, y_o : B)$	$(x_o : B)$	$(x_o : B)$

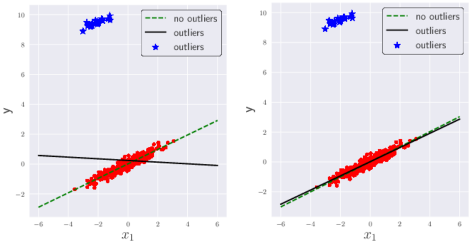
Mathematical intuition: For outlier z with $p(z|\theta) \approx 0$:

$$l_\beta(z) = \frac{\beta+1}{\beta} p(z|\theta)^\beta - \int p(x|\theta)^{1+\beta} dx \quad \text{stays bounded!}$$

Toy Experiment: Linear Regression with Outliers

Experimental Setup (Section 5.1):

- 2D toy dataset with linear relationship: $y = w_0 + w_1 x_1 + w_2 x_2 + \epsilon$
- Artificial input-related outliers added (measurement errors)
- Model: Gaussian likelihood with mean-field variational inference
- Compared: Ordinary VI vs Proposed β-VI ($\beta = 0.1$)



(a) Ordinary VI with outliers (b) Proposed VI ($\beta = 0.1$) with outliers

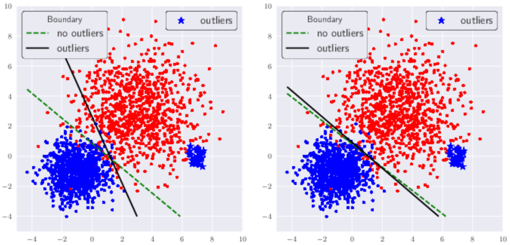
Figure 1: Linear regression. Predictive distributions are derived by variational inference (VI).

Authors' Results:

× **Ordinary VI:** predictive distribution heavily affected by outliers
  - Regression line pulled toward outliers
  - Wide uncertainty bands

✓ **Proposed β-VI:** predictive distribution less affected by outliers

Toy Experiment: Logistic Regression (Section 5.1)



(a) Ordinary VI with outliers

(b) Proposed VI (β=0.4) with outliers

Figure 2: Boundaries of logistic regression using ordinary VI and the proposed method

Task: Binary classification

Outliers: Wrongly labeled points

Results:

- Ordinary VI: boundary heavily distorted
- β-VI: boundary stable, clean separation

Mechanism:

$$p(y_{\text{wrong}}|x,\theta)^{\beta} \to 0$$

Mislabeled points get near-zero weight

Conclusion: Visual confirmation of robustness to output-related outliers (label noise) — critical for real-world classification tasks with annotation errors